

The residual RMS bound, revised: the envelope-witness chain

figures `fig_bound_final_sim`, `fig_bound_final_hw`; scripts in the Reproduction section

Symbols

τ, τ_{end}	time since onset; window end	x	$= C_2 \tau_{\text{end}}$, dimensionless window
y_i	measured rate at τ_i	N_s	number of samples in the window
C_1, C_2	nominal amplitude $K\dot{M}$ and exponent	$K, \dot{M}$	gain $1/Wz_{CoM}$; ramp rate
W, z_{CoM}, J_P	weight, CoM height, pivot inertia	Wz_{CoM}	$= J_P C_2^2$ (the identity)
ω_{nom}	nominal rate $C_1(\cosh C_2 \tau - 1)$	t_{on}	realised onset time
$e_\omega(\tau)$	true rate minus the nominal	b, n_i	gyro bias; noise sample
$\hat{h}(\cdot; \theta)$	fitted family member	θ	(C_1^*, C_2^*, t^*, C) , all free
r_i	residual $y_i - \hat{h}(\tau_i; \hat{\theta})$	$E(\tau)$	envelope $\bar{\rho} K C_2 \sinh(C_2 \tau)$
$\rho(s)$	deviation forcing (small-angle + GE)	$\bar{\rho}$	window bound on $ \rho $, eq. (12)
$B(x)$	$\frac{1}{4} \sinh 2x - \frac{x}{2}$ ($= \int_0^x \sinh^2 u \, du$)	φ_{max}	design tilt cap (5° sim, 8° hw)
l_{arm}, β_M	pivot arm; GE coefficient	ΔM_{win}	in-window moment sweep $\dot{M}x/C_2$
n_{hi}	residual content above 5 Hz	κ_b	quiet-segment band ratio (1.31)
σ_n	declared sim gyro std, 2.450×10^{-4} rad/s		

1 The claim

For every run of both campaigns, with every quantity on the right computable before the experiment except the hardware vibration amplitude n_{hi} (read from the run's own out-of-band content),

$$\boxed{\text{RMS}(r) \leq \text{RMS}(e_\omega) + \text{RMS}(n) \leq \bar{\rho} K C_2 \sqrt{\frac{B(x)}{x}} + N.} \quad (1)$$

$\text{RMS}(r)$ is the full-window RMS of the deployed PNLS fit's residual — the fit exactly as it runs in the pipeline, both shape constants and the onset free, no calibration anywhere. N is the noise term of Sec. 6: negligible ($\sigma_n = 0.014^\circ/\text{s}$) in simulation, the measured vibration term $n_{\text{hi}}\sqrt{1 + \kappa_b^2}$ on hardware. The two inequalities have entirely different characters and are proved separately: the first is an optimisation triviality (Sec. 3), the second is the physics (Sec. 4). Validation: 301/308 simulation runs and 134/140 hardware runs inside, with every exceedance located and explained (Sec. 7).

2 The measurement and the family

In the excitation window the record is

$$y_i = \omega_{\text{nom}}(\tau_i) + e_\omega(\tau_i) + b + n_i, \quad \omega_{\text{nom}}(\tau) = \begin{cases} C_1 (\cosh(C_2 \tau) - 1), & \tau \geq 0, \\ 0, & \tau < 0, \end{cases} \quad (2)$$

with τ counted from the *realised* onset t_{on} : the instant the restoring moment actually reverses at the vehicle, wherever the actuation chain put it relative to the commanded ramp clock. The deviation $e_\omega = y - b - n - \omega_{\text{nom}}$ collects everything the small-angle model omits — trigonometric remainders, ground effect — and vanishes identically before the onset.

The estimator fits the four-parameter family

$$\hat{h}(\tau; \theta) = C_1^* (\cosh(C_2^*(\tau - t^*)) - 1) \mathbf{1}[\tau \geq t^*] + C, \quad \theta = (C_1^*, C_2^*, t^*, C), \quad (3)$$

by least squares over the full window, $\hat{\theta} = \arg \min_{\theta} \sum_i (y_i - \hat{h}(\tau_i; \theta))^2$, and $r_i = y_i - \hat{h}(\tau_i; \hat{\theta})$.

3 The first inequality: why the bound lands on e_ω

The point of this section is one sentence: *the nominal solution is itself a member of the fitted family, so the fit cannot lose to it.* Take the particular parameter vector

$$\theta_0 = (C_1, C_2, t_{\text{on}}, b) \implies \hat{h}(\tau; \theta_0) = \omega_{\text{nom}}(\tau) + b. \quad (4)$$

Nothing in (3) forbids this choice: the amplitude slot takes C_1 , the exponent slot C_2 , the onset slot the realised t_{on} , the baseline slot the bias. θ_0 is called the *witness*: not the minimiser, not a good fit, merely one admissible competitor whose residual can be computed exactly. A minimum is bounded by the value at any feasible point, so

$$\sum_i r_i^2 = \min_\theta \sum_i (y_i - \hat{h}(\tau_i; \theta))^2 \leq \sum_i (y_i - \hat{h}(\tau_i; \theta_0))^2 = \sum_i (e_\omega(\tau_i) + n_i)^2, \quad (5)$$

where the last equality is (2) minus (4). Dividing by N_s and taking the square root, then splitting by the triangle inequality in the sample ℓ^2 norm,

$$\text{RMS}(r) \leq \text{RMS}(e_\omega + n) \leq \text{RMS}(e_\omega) + \text{RMS}(n). \quad (6)$$

That is the whole proof. Four remarks fix its scope.

(i) *What is and is not claimed about the minimiser.* (5) does not say the fit finds the nominal, or anything close to it: the minimiser wanders wherever the data pulls it (the amplitude–onset ridge of the earlier analysis is real). The inequality runs one way only — the fit does *at least as well* as the nominal — and one way is all a residual *upper* bound needs. The measured residuals sitting far below the cap (median used fraction 0.47–0.48 on both campaigns) is the minimiser exploiting freedoms the witness does not use. Geometrically: the measured curve lies between the nominal and the nominal-plus-envelope, and the fitted curve, free to move, lands at least as close to the measurement as the lower edge of that corridor — so the fit’s residual is capped by the corridor’s width, which is exactly $\text{RMS}(E)$.

(ii) *No admissibility condition.* The earlier draft’s witness was the nominal *plus* a $\beta \sinh$ correction, folded back into the family through a time shift — legitimate only while $|\beta| < C_1$, which dragged the dimensionless ratio $u = \bar{\rho} C_2 / M$ and a lower ramp-rate limit into the certificate. The witness (4) is the nominal itself, a family member by construction, unconditionally. The $u < 1$ discussion survives only where it always belonged: whether a *fitted* branch has a physical onset to read off, not whether the residual is capped.

(iii) *The onset freedom absorbs pure delay.* Every transport and actuation lag between the commanded ramp and the vehicle shifts t_{on} , and nothing else: the tip-over dynamics start when the moment balance reverses at the vehicle, and carry no memory of the command clock. Because t^* is free in (3), the witness uses the realised onset, and e_ω — defined *against the realised-onset nominal* — obeys the deviation dynamics of Sec. 4 regardless of the delay’s size. This is why no latency channel appears anywhere in (1): a pure delay is invisible to the chain. (A delay that *distorts* the ramp within the window is not pure and lands in ρ ; the measured achieved-moment ramps are straight to 3%, and the antisymmetric part cancels in the direction-pair half-sum.)

(iv) *Full window against post-onset window.* $\text{RMS}(r)$ is reported over the full window (pre-onset segment included), while the envelope of Sec. 4 lives on the post-onset segment. The directions compose correctly: $e_\omega \equiv 0$ and $\hat{h}(\cdot; \theta_0) = b$ before the onset, so the pre-onset segment contributes only noise to the witness’s residual, and

$$\text{RMS}_{\text{full}}(e_\omega) = \sqrt{\frac{T_{\text{post}}}{T_{\text{full}}}} \text{RMS}_{\text{post}}(e_\omega) \leq \text{RMS}_{\text{post}}(e_\omega), \quad (7)$$

so bounding the post-onset RMS bounds the reported quantity a fortiori.

4 The second inequality: the envelope and its window RMS

The deviation obeys the manuscript’s forced tip-over equation ((89) there): with e_φ the tilt deviation, $e_\omega = \dot{e}_\varphi$,

$$J_P \ddot{e}_\varphi = W z_{CoM} e_\varphi + \rho(\tau), \quad e_\varphi(0) = \dot{e}_\varphi(0) = 0, \quad |\rho(s)| \leq \bar{\rho} \text{ on } [0, \tau_{\text{end}}]. \quad (8)$$

Duhamel with the hyperbolic kernel and one differentiation gives

$$e_\omega(\tau) = \frac{1}{J_P} \int_0^\tau \cosh(C_2(\tau - s)) \rho(s) ds; \quad (9)$$

bounding ρ by $\bar{\rho}$ and integrating the kernel,

$$|e_\omega(\tau)| \leq \frac{\bar{\rho}}{J_P C_2} \sinh(C_2 \tau) = \bar{\rho} K C_2 \sinh(C_2 \tau) \triangleq E(\tau), \quad (10)$$

the manuscript's envelope (90); the identity $K C_2 = 1/(J_P C_2)$ follows from $K = 1/W z_{CoM}$ and $W z_{CoM} = J_P C_2^2$. A pointwise bound gives an RMS bound by integration: with $u = C_2 \tau$ and $x = C_2 \tau_{\text{end}}$,

$$\text{RMS}_{\text{post}}(e_\omega) \leq \bar{\rho} K C_2 \sqrt{\frac{1}{x} \int_0^x \sinh^2 u \, du} = \bar{\rho} K C_2 \sqrt{\frac{B(x)}{x}}, \quad B(x) = \frac{\sinh 2x}{4} - \frac{x}{2}, \quad (11)$$

B by the double-angle identity $\sinh^2 u = \frac{1}{2}(\cosh 2u - 1)$. The factor $\sqrt{B(x)/x}$ is simply the window RMS of the shape $\sinh(C_2 \tau)$: $\approx x/\sqrt{3}$ for small windows, $\approx e^x/(2\sqrt{2}x)$ for long ones — the price of the instability's growing mode, paid honestly rather than absorbed by a fitted term.

5 The forcing bound $\bar{\rho}$ and the design tilt cap

The channels are the manuscript's (91), window-averaged with its reduction factors $R_\varphi = 1/7$ and $R_{GE} = 1/5$:

$$\bar{\rho} = R_\varphi \cdot \frac{1}{2} W l_{\text{arm}} \varphi_{\text{max}}^2 + R_{GE} \cdot |\beta_M| \Delta M_{\text{win}} \varphi_{\text{max}}, \quad (12)$$

with the window quantities set a priori per ramp rate — the nominal tilt reaches the cap when

$$\sinh x - x = \frac{\varphi_{\text{max}} W z_{CoM} C_2}{\dot{M}}, \quad \Delta M_{\text{win}} = \frac{\dot{M} x}{C_2}, \quad (13)$$

solved once per rate before any run. The simulator models no ground effect, so its $\bar{\rho}$ carries the first term alone, with the per-case arm $l_{\text{arm}} = l_p + p_{\text{off}}$, $z_{CoM} = 0.272$ m, and $J_P = J^{\text{CAD}} + m(z_{CoM}^2 + l_p^2)$; hardware uses the design box of Sec. VI-D ($z_{CoM} = 0.30$ m, arms 0.160/0.130 m, $|\beta_M| = 0.0345$).

φ_{max} is not a tuning knob: (10) holds on the window only while the trajectory stays inside the box on which (12) bounds ρ , so *the cap must dominate the realised in-window tilt*, and its value is set by each campaign's own geometry. In simulation the gate stops the window near 4° realised tilt: $\varphi_{\text{max}} = 5^\circ$. On hardware the slightly uneven ground of the test site lets the in-window tilt rise reach 7.66° (median 4.82° , quaternion-measured across all 140 runs): $\varphi_{\text{max}} = 8^\circ$. Every appearance of φ_{max} in (12)–(13) is monotone, so a cap that dominates the realised tilt only over-covers.

6 The noise term N

$\text{RMS}(n)$ in (6) is whatever the sensor adds that the physics does not produce.

Simulation. The declared sensor model: additive Gaussian rate noise of standard deviation $\sigma_n = 2.450 \times 10^{-4}$ rad/s = 0.014°/s. This sits an order of magnitude below the smallest envelope value on the campaign, so the validation figure carries the envelope alone; including σ_n changes no count.

Hardware. The vehicle vibrates under its running rotors, and no pre-experiment model of that channel exists, so its amplitude is the one measured input of the chain: per run, n_{hi} is the RMS of the residual's content above 5 Hz — a band the family cannot occupy, since its fastest structural scale is $C_2/2\pi \approx 0.8$ Hz — extended to the full band by the campaign constant $\kappa_b = 1.31$, the below-5-Hz to above-5-Hz vibration ratio measured on quiet segments outside every excitation window:

$$N = n_{\text{hi}} \sqrt{1 + \kappa_b^2}. \quad (14)$$

7 Validation, both campaigns

Per run: the deployed free fit's full-window residual RMS against the cap of (1). No calibration enters on either side.

campaign	runs	inside	worst used fraction	median	p90
simulation ($\varphi_{\text{max}} = 5^\circ$, envelope alone)	308	301	2.30	0.48	0.73
hardware ($\varphi_{\text{max}} = 8^\circ$, envelope + (14))	140	134	1.65	0.47	0.81

The simulation exceedances, all seven at the two fastest ramps: three marginal (1.04–1.07: $S1/M_y$, $S2/M_x$, $S9/M_x$ at 1.20 N m/s) and four substantial (1.23–2.30), every one of the four on $S9/M_y$ or $S11/M_y$ — the injected offsets (32, 32) and (38, 14) mm sit at or beyond the design rectangle whose arm the $\bar{\rho}$ of (12) assumes, so these runs operate outside the box the bound is issued for.

The hardware exceedances, six runs between 0.65 and 1.20 N m/s (worst used fraction 1.65), carry disturbance content outside the modelled channels: (12) budgets the small-angle remainder and ground effect, and (14) the rotor-induced vibration, but an outdoor vehicle on real ground also meets unmodelled disturbances — gusts, contact irregularities at the pivot — that enter ρ with no term reserved for them. Six runs in 140 carrying a visible dose of them, at the fast ramps where the window is shortest and a single transient weighs most in the RMS, is the expected signature; the remaining 134 sit inside with the used-fraction distribution above.

The two campaigns validate *one* construction: the same chain (1), the same witness, the same envelope, the same design-box $\bar{\rho}$. The panels differ in exactly the channels the platforms differ in — the simulator declares its noise and models no ground effect, so its cap is the envelope alone; the vehicle adds ground effect (inside $\bar{\rho}$), vibration (the term (14)), and, on the six runs above, disturbances no pre-experiment budget can name. Figures `fig_bound_final_sim` and `fig_bound_final_hw` show every run.

8 What remains of the shaped decomposition

The earlier draft of this document ran the bound through the decomposition $e_\omega = \beta \sinh(C_2\tau) + \delta e_\omega$ with $\beta = e_\omega(\tau_{\text{end}})/\sinh(x)$ (manuscript (95)–(97)), capping δe_ω by the Dirichlet comparison constants $1/12$ and $1/(2e)$ (manuscript (107)). Measured directly — pinned-theory fit, β from the definition, δe_ω pointwise — that cap does not cover: on the simulation campaign 0 of 301 runs sit inside it (realised δe_ω RMS 2–3× the cap), and on hardware only the below-5-Hz band clears it, marginally (101/139). The decomposition is an identity and survives as one; what fails is the premise that the remainder after the end-matched sinh subtraction is dominated by the in-family forcing response — the subtraction is exact only at the two window ends, and the out-of-family content between them is amplified by the same sinh growth the envelope charges for.

Its residual role is therefore qualitative: it explains *why the measured residuals are flat in the ramp rate* (the fit’s onset freedom absorbs the growing mode’s dominant multiple, which is the tangent direction of the time shift), while the quantitative certificate is (1). For the manuscript this means: state (1) as the headline bound with the two-line proof of Sec. 3; keep (95)–(97) as the flatness explanation, dropping the sample index i from them (they are function-level identities; i belongs only to (92) and the least-squares objective); and demote (98)–(107) to a remark scoped to what it is — a bound on the forcing response, not on the measured remainder.

9 Reproduction

<code>analysis/sim_envelope_witness.py</code>	simulation sweep → <code>docs/sim_env_witness_runs.csv</code>
<code>analysis/hw_envelope_witness.py</code>	hardware sweep → <code>docs/hw_env_noise_runs.csv</code>
<code>analysis/bound_final_figs.py</code>	caps of (1), both figures, the counts of Sec. 7
<code>analysis/de_measure.py</code>	the direct δe_ω measurement of Sec. 8

The residual in every script is the pipeline’s own stage-1 call, `extract_piecewise(model='cosh', cosh_c2=None, ramp_gain=None)` — both constants and the onset free, the full window, no calibration.